# TMA1001/TCX1101 Introductory Mathematics

Tutorial 4

# Question 1(a)

The angle between two vectors $\mathbf{a}$ and $\mathbf{b}$ is $120°$. If $\|\mathbf{a}\| = 3$ and $\|\mathbf{b} - \mathbf{a}\| = 7$, find $\|\mathbf{b}\|$.

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The angle between two vectors $\mathbf{a}$ and $\mathbf{b}$ is $120°$. If $\|\mathbf{a}\| = 3$ and $\|\mathbf{b} - \mathbf{a}\| = 7$, find $\|\mathbf{b}\|$.

Let $x = \|\mathbf{b}\|$. By the law of cosine:

$$\|\mathbf{b} - \mathbf{a}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\cos 120° \|\mathbf{a}\| \, \|\mathbf{b}\|,$$

i.e., $7^2 = 3^2 + x^2 + 3x$. Then $0 = x^2 + 3x - 40 = (x+8)(x-5)$, which implies $x = -8$ (rejected, since $x = \|\mathbf{b}\| > 0$) or $x = 5$. Hence, $\|\mathbf{b}\| = 5$.

# Question 1(b)

The angle between two vectors $\mathbf{a}$ and $\mathbf{b}$ is $120^\circ$. If $\|\mathbf{a}\| = 3$ and $\|\mathbf{b} - \mathbf{a}\| = 7$, find $\|\mathbf{a} + \mathbf{b}\|$.

# Question 1(b)

The angle between two vectors $\mathbf{a}$ and $\mathbf{b}$ is $120^\circ$. If $\|\mathbf{a}\| = 3$ and $\|\mathbf{b} - \mathbf{a}\| = 7$, find $\|\mathbf{a} + \mathbf{b}\|$.

Compute directly

$$\|\mathbf{a} + \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 + 2\,\mathbf{a} \cdot \mathbf{b} = 9 + 25 + 2(3 \cdot 5 \cos 120^\circ) = 34 - 15 = 19,$$

so, $\|\mathbf{a} + \mathbf{b}\| = \sqrt{19}$.

# Question 2(a)

Let $A(0,3,4)$, $B(-2,p,3)$, $C(q,1,3)$ and $D(4,7,r)$ be points in $\mathbb{R}^3$. Find the value of $p$ for which $\|\overrightarrow{AB}\| = 3$.

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Let $A(0,3,4)$, $B(-2,p,3)$, $C(q,1,3)$ and $D(4,7,r)$ be points in $\mathbb{R}^3$. Find the value of $p$ for which $\|\overrightarrow{AB}\| = 3$.

$$\overrightarrow{AB} = \begin{pmatrix} -2 \\ p-3 \\ -1 \end{pmatrix}. \quad \text{Then} \quad \|\overrightarrow{AB}\|^2 = (-2)^2 + (p-3)^2 + (-1)^2 = 5 + (p-3)^2.$$

$$3^2 = 9 = \|\overrightarrow{AB}\|^2 \quad \Rightarrow \quad (p-3)^2 = 4 \quad \Rightarrow \quad p = 1 \text{ or } p = 5.$$

# Question 2(b)

Let $A(0, 3, 4)$, $B(-2, p, 3)$, $C(q, 1, 3)$ and $D(4, 7, r)$ be points in $\mathbb{R}^3$. Find the values of $p$ and $r$ for which $A, B$ and $D$ are collinear, i.e., lie on the same line.

# Question 2(b)

Let $A(0,3,4)$, $B(-2,p,3)$, $C(q,1,3)$ and $D(4,7,r)$ be points in $\mathbb{R}^3$. Find the values of $p$ and $r$ for which $A, B$ and $D$ are collinear, i.e., lie on the same line.

If the points are collinear, then $\overrightarrow{AB} = \lambda\overrightarrow{AD}$ for some $\lambda \in \mathbb{R}$.

$$\begin{pmatrix} -2 \\ p-3 \\ -1 \end{pmatrix} = \lambda \begin{pmatrix} 4 \\ 4 \\ r-4 \end{pmatrix} = \begin{pmatrix} 4\lambda \\ 4\lambda \\ \lambda(r-4) \end{pmatrix} \implies \begin{cases} -2 = 4\lambda \\ p-3 = 4\lambda \\ -1 = \lambda(r-4) \end{cases} \implies \begin{cases} \lambda = -\frac{1}{2} \\ p = 1 \\ r = 6 \end{cases}$$

$$\lambda = -\frac{1}{2}, \quad p = 1, \quad r = 6.$$

# Question 2(c)

Let $A(0,3,4)$, $B(-2,p,3)$, $C(q,1,3)$ and $D(4,7,r)$ be points in $\mathbb{R}^3$. Find the value of $q$ for which $\overrightarrow{AC} \perp \overrightarrow{OC}$, where $O$ is the origin.

# Question 2(c)

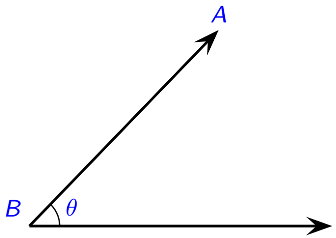
Let $A(0,3,4)$, $B(-2,p,3)$, $C(q,1,3)$ and $D(4,7,r)$ be points in $\mathbb{R}^3$. Find the value of $q$ for which $\overrightarrow{AC} \perp \overrightarrow{OC}$, where $O$ is the origin.

$\overrightarrow{AC} \perp \overrightarrow{OC}$ if $\cos\theta = 0$, where $\theta$ is the angle between $\overrightarrow{AC}$ and $\overrightarrow{OC}$. From the definition of angle $\cos\theta = \dfrac{\overrightarrow{AC} \cdot \overrightarrow{OC}}{\|\overrightarrow{AC}\|\|\overrightarrow{OC}\|}$, this means that $\overrightarrow{AC} \cdot \overrightarrow{OC} = 0$.

$$0 = \begin{pmatrix} q \\ -2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} q \\ 1 \\ 3 \end{pmatrix} = q^2 - 5 \implies q = \pm\sqrt{5}.$$

# Question 2(d)

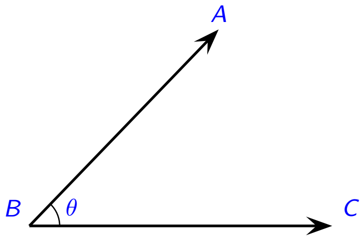
Let $A(0, 3, 4)$, $B(-2, p, 3)$, $C(q, 1, 3)$ and $D(4, 7, r)$ be points in $\mathbb{R}^3$. Find $\angle ABC$ if $p = 1$ and $q = 2$.



# Question 2(d)

Let $A(0,3,4)$, $B(-2,p,3)$, $C(q,1,3)$ and $D(4,7,r)$ be points in $\mathbb{R}^3$. Find $\angle ABC$ if $p=1$ and $q=2$.

$$\cos\angle ABC = \frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{\|\overrightarrow{BA}\|\,\|\overrightarrow{BC}\|} = \frac{\begin{pmatrix}2\\2\\1\end{pmatrix}\cdot\begin{pmatrix}4\\0\\0\end{pmatrix}}{\left\|\begin{pmatrix}2\\2\\1\end{pmatrix}\right\|\left\|\begin{pmatrix}4\\0\\0\end{pmatrix}\right\|} = \frac{8}{\sqrt{9}\cdot 4} = \frac{2}{3}.$$




Therefore, $\angle ABC = \cos^{-1}\left(\frac{2}{3}\right) \approx 0.8411$ radians $\approx 48.2^\circ$.

# Question 3(a)

Find the unit vector in the direction of $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

$$\frac{1}{\left\| \begin{pmatrix} -4 \\ 3 \end{pmatrix} \right\|} \begin{pmatrix} -4 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{25}} \begin{pmatrix} -4 \\ 3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -4 \\ 3 \end{pmatrix}.$$

# Question 3(b)

Find two vectors which have length 34 and are parallel to $\begin{pmatrix} 4 \\ -\frac{15}{2} \end{pmatrix}$.

https://www.geogebra.org/calculator

# Question 3(b)

Find two vectors which have length 34 and are parallel to $\begin{pmatrix} 4 \\ -\frac{15}{2} \end{pmatrix}$.

`https://www.geogebra.org/calculator`

First find the unit vector parallel to $\begin{pmatrix} 4 \\ -\frac{15}{2} \end{pmatrix}$.

$$\frac{1}{\left\| \begin{pmatrix} 4 \\ -\frac{15}{2} \end{pmatrix} \right\|} \begin{pmatrix} 4 \\ -\frac{15}{2} \end{pmatrix} = \frac{1}{\sqrt{16 + \frac{225}{4}}} \begin{pmatrix} 4 \\ -\frac{15}{2} \end{pmatrix} = \cdots = \frac{1}{17} \begin{pmatrix} 8 \\ -15 \end{pmatrix}.$$

Vector of magnitude 34 in the direction of $\frac{1}{17}\begin{pmatrix} 8 \\ -15 \end{pmatrix}$ is $\frac{34}{17}\begin{pmatrix} 8 \\ -15 \end{pmatrix} = \begin{pmatrix} 16 \\ -30 \end{pmatrix}$, and in the opposite direction $\begin{pmatrix} -16 \\ 30 \end{pmatrix}$.

# Question 4(a)

The position vectors of points $A$, $B$ and $C$ are

$$\mathbf{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Show that $\triangle ABC$ is an isosceles triangle.

# Question 4(a)

The position vectors of points $A$, $B$ and $C$ are

$$\mathbf{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Show that $\triangle ABC$ is an isosceles triangle.

Show that 2 sides has same length.

$$\left\|\overrightarrow{AB}\right\| = \|\mathbf{b} - \mathbf{a}\| = \left\|\begin{pmatrix} -4 \\ 3 \end{pmatrix}\right\| = \sqrt{25} = 5.$$

$$\left\|\overrightarrow{AC}\right\| = \|\mathbf{c} - \mathbf{a}\| = \left\|\begin{pmatrix} -3 \\ 4 \end{pmatrix}\right\| = \sqrt{25} = 5.$$

# Question 4(b)

The position vectors of points $A, B$ and $C$ are

$$\mathbf{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Find $\angle BAC$, and hence find the area of $\triangle ABC$.

# Question 4(b)

The position vectors of points $A$, $B$ and $C$ are

$$\mathbf{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Find $\angle BAC$, and hence find the area of $\triangle ABC$.

Area of $\triangle ABC = \frac{1}{2} \cdot \|\overrightarrow{AB}\| \cdot \|\overrightarrow{AC}\| \cdot \sin\angle BAC$.

$$\cos\angle BAC = \frac{\overrightarrow{AB} \cdot \overrightarrow{AC}}{\|\overrightarrow{AB}\|\,\|\overrightarrow{AC}\|} = \frac{\begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 4 \end{pmatrix}}{\left\|\begin{pmatrix} -4 \\ 3 \end{pmatrix}\right\| \left\|\begin{pmatrix} -3 \\ 4 \end{pmatrix}\right\|} = \frac{24}{25} \implies \sin\angle BAC = \sqrt{1 - \left(\frac{24}{25}\right)^2} = \frac{7}{25}.$$

$$\frac{1}{2} \cdot \|\overrightarrow{AB}\| \cdot \|\overrightarrow{AC}\| \cdot \sin\angle BAC = \frac{1}{2} \cdot 5 \cdot 5 \cdot \frac{7}{25} = \frac{7}{2}.$$

# Question 5(a)

Compute $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$.

# Question 5(a)

Compute $\begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}3\\2\\1\end{pmatrix}$.

$$\left|\begin{array}{ccc|cc} x & y & z & x & y \\ 1 & 2 & 3 & 1 & 2 \\ 3 & 2 & 1 & 3 & 2 \end{array}\right|$$

$$\begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}3\\2\\1\end{pmatrix} = \begin{pmatrix}2\cdot 1 - 3\cdot 2\\3\cdot 3 - 1\cdot 1\\1\cdot 2 - 2\cdot 3\end{pmatrix} = \begin{pmatrix}-4\\8\\-4\end{pmatrix}.$$

# Question 5(b)

Compute $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \left[ \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right]$.

# Question 5(b)

Compute $\begin{pmatrix}1\\2\\3\end{pmatrix} \times \left[\begin{pmatrix}3\\2\\1\end{pmatrix} \times \begin{pmatrix}1\\1\\1\end{pmatrix}\right]$.

$$\begin{aligned}
\begin{pmatrix}1\\2\\3\end{pmatrix} \times \left[\begin{pmatrix}3\\2\\1\end{pmatrix} \times \begin{pmatrix}1\\1\\1\end{pmatrix}\right] &= \begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}2\cdot 1 - 1\cdot 1\\1\cdot 1 - 3\cdot 1\\3\cdot 1 - 2\cdot 1\end{pmatrix}\\
&= \begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}1\\-2\\1\end{pmatrix} = \begin{pmatrix}2\cdot 1 - 3\cdot(-2)\\3\cdot 1 - 1\cdot 1\\1\cdot(-2) - 2\cdot 1\end{pmatrix}\\
&= \begin{pmatrix}8\\2\\-4\end{pmatrix}.
\end{aligned}$$

# Question 6

Let $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$, and $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$. Find a unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

## Question 6

Let $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$, and $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$. Find a unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

To find a vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, we compute the cross product

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} (-1)(1) - (3)(4) \\ (3)(-1) - (2)(1) \\ (2)(4) - (-1)(-1) \end{pmatrix} = \begin{pmatrix} -1 - 12 \\ -3 - 2 \\ 8 - 1 \end{pmatrix} = \begin{pmatrix} -13 \\ -5 \\ 7 \end{pmatrix}.$$

Now normalize:

$$\frac{1}{\left\| \begin{pmatrix} -13 \\ -5 \\ 7 \end{pmatrix} \right\|} \begin{pmatrix} -13 \\ -5 \\ 7 \end{pmatrix} = \frac{1}{\sqrt{243}} \begin{pmatrix} -13 \\ -5 \\ 7 \end{pmatrix}.$$

# Question 7(a)

Two lines $L_1$ and $L_2$ have vector equations given respectively by

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}, \qquad \text{and} \qquad \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mu \in \mathbb{R}.$$

Show that $L_1$ and $L_2$ intersect, and find the point of intersection.

# Question 7(a)

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Show that $L_1$ and $L_2$ intersect, and find the point of intersection.

Let the two expressions for $\mathbf{r}$ be equal:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \implies \begin{cases} 1 + 2\lambda = 4 + \mu, \\ 1 + \lambda = 1, \\ 1 + \lambda = 10 + 3\mu \end{cases} \implies \lambda = 0, \mu = 3.$$

# Question 7(b)

Two lines $L_1$ and $L_2$ have vector equations given respectively by

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}, \qquad \text{and} \qquad \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mu \in \mathbb{R}.$$

Find the acute angle between $L_1$ and $L_2$.

# Question 7(b)

Two lines $L_1$ and $L_2$ have vector equations given respectively by

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}, \qquad \text{and} \qquad \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mu \in \mathbb{R}.$$

Find the acute angle between $L_1$ and $L_2$.

Angle between the lines is given by $\cos\theta = \dfrac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{\|\mathbf{d}_1\| \, \|\mathbf{d}_2\|}$, where $\mathbf{d}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$.

$$\cos\theta = \frac{\left| \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \right|}{\left\| \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \right\| \, \left\| \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \right\|} = \frac{5}{\sqrt{6 \cdot 10}} = \frac{\sqrt{15}}{6} \implies \theta = \cos^{-1}\left( \frac{\sqrt{15}}{6} \right).$$

# Question 7(c)

Two lines $L_1$ and $L_2$ have vector equations given respectively by

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}, \qquad \text{and} \qquad \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mu \in \mathbb{R}.$$

Show that the point $A(3, 3, 7)$ is not on $L_1$, and determine the projection of $A$ onto $L_1$.

# Question 7(c)

Two lines $L_1$ and $L_2$ have vector equations given respectively by

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}, \qquad \text{and} \qquad \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mu \in \mathbb{R}.$$

Show that the point $A(3,3,7)$ is not on $L_1$, and determine the projection of $A$ onto $L_1$.

Point $A$ is on $L_1$ if there is a $\lambda$ such that

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 7 \end{pmatrix} \implies \begin{cases} 1 + 2\lambda = 3 & \Rightarrow \quad \lambda = 1, \\ 1 + \lambda = 3 & \Rightarrow \quad \lambda = 2, \\ 1 + \lambda = 7 & \Rightarrow \quad \lambda = 6. \end{cases}$$

he equations are inconsistent, so $A$ is not on $L_1$.

# Question 7(c)

Now, let $\mathbf{p}$ be the foot of the perpendicular from $A$ to $L_1$, that is, for some $\lambda$,

$$\mathbf{p} = \begin{pmatrix}1\\1\\1\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\1\end{pmatrix} \implies \overrightarrow{AP} = \mathbf{p} - \begin{pmatrix}3\\3\\7\end{pmatrix} = \begin{pmatrix}-2+2\lambda\\-2+\lambda\\-6+\lambda\end{pmatrix}.$$

Since $\overrightarrow{AP} \perp \mathbf{d}_1$, we have:

$$\begin{pmatrix}-2+2\lambda\\-2+\lambda\\-6+\lambda\end{pmatrix}\cdot\begin{pmatrix}2\\1\\1\end{pmatrix} = 0 \implies (-2+2\lambda)(2)+(-2+\lambda)(1)+(-6+\lambda)(1) = 0 \implies \lambda = 2.$$

So, $\mathbf{p} = \begin{pmatrix}1\\1\\1\end{pmatrix} + 2\begin{pmatrix}2\\1\\1\end{pmatrix} = \begin{pmatrix}5\\3\\3\end{pmatrix}$.

# Question 8(a)

The point $P$ has position vector $\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$ and the point $Q$ has position vector $\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$. Find a vector equation of the line $L$ passing through $P$ and $Q$.

# Question 8(a)

The point $P$ has position vector $\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$ and the point $Q$ has position vector $\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$. Find a vector equation of the line $L$ passing through $P$ and $Q$.

A direction vector of $L$ is

$$\overrightarrow{PQ} = \mathbf{q} - \mathbf{p} = \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}.$$

Then a vector equation of the line is

$$\mathbf{r} = \mathbf{p} + \lambda \overrightarrow{PQ} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

# Question 8(b)

The point $P$ has position vector $\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$ and the point $Q$ has position vector $\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$. Find two points on $L$ which are $\sqrt{35}$ units from the origin $O$.

# Question 8(b)

The point $P$ has position vector $\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$ and the point $Q$ has position vector $\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$. Find two points on $L$ which are $\sqrt{35}$ units from the origin $O$.

$$\|\mathbf{r}\| = \sqrt{(4-\lambda)^2 + (1-2\lambda)^2} = \sqrt{35} \implies 6\lambda^2 - 12\lambda + 17 = 35 \implies (\lambda+3)(\lambda-1) = 0.$$

$$\lambda = -3 \quad \implies \quad \mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + (-3)\begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 7 \end{pmatrix}.$$

$$\lambda = 1 \quad \implies \quad \mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}.$$

# Question 9(a)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find the position vectors of $C$ and $D$.

# Question 9(a)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find the position vectors of $C$ and $D$.

$$\mathbf{c} = 2\,\mathbf{a} = 2\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix} = \begin{pmatrix} 4 \\ 18 \\ -12 \end{pmatrix}.$$

$$\overrightarrow{AB} = \begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ 12 \end{pmatrix}, \quad \mathbf{d} = \mathbf{a} + \frac{1}{2}\overrightarrow{AB} = \begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 4 \\ -6 \\ 12 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 0 \end{pmatrix}.$$

# Question 9(b)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find a vector equation of the line $L$ passing through $C$ and $D$.

# Question 9(b)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find a vector equation of the line $L$ passing through $C$ and $D$.

$$\overrightarrow{CD} = \mathbf{d} - \mathbf{c} = \begin{pmatrix} 4 \\ 6 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 18 \\ -12 \end{pmatrix} = \begin{pmatrix} 0 \\ -12 \\ 12 \end{pmatrix}.$$

Hence, the line $L$ has equation:

$$\mathbf{r} = \mathbf{c} + \lambda\overrightarrow{CD} = \begin{pmatrix} 4 \\ 18 \\ -12 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -12 \\ 12 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

# Question 9(c)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find the point at which $L$ intersects the line passing through $O$ and $B$.

# Question 9(c)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find the point at which $L$ intersects the line passing through $O$ and $B$.

The line that passes through $O$ and $B$ is

$$\mathbf{r} = O + \mu\overrightarrow{OB} = \mu \begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix} = \mu \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \quad \mu \in \mathbb{R}.$$

Now solve for point of intersection:

$$\begin{pmatrix} 4 \\ 18 - 12\lambda \\ -12 + 12\lambda \end{pmatrix} = \begin{pmatrix} 2\mu \\ \mu \\ 2\mu \end{pmatrix} \Longrightarrow \begin{cases} 4 = 2\mu & \Rightarrow \quad \mu = 2, \\ 18 - 12\lambda = 2 & \Rightarrow \quad \lambda = \dfrac{4}{3}, \\ -12 + 12\lambda = 4 & \Rightarrow \quad \lambda = \dfrac{4}{3}. \end{cases}$$

# Question 9(c)

Substituting $\mu = 2$ into the line through $O$ and $B$:

$$\mathbf{r} = \begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix}.$$

# Question 9(d)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$.

# Question 9(d)

The point $A$ has position vector $\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}$, and the point $B$ has position vector $\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}$. The point $C$ is such that $\overrightarrow{OC} = 2\,\overrightarrow{OA}$, and $D$ is the midpoint of line segment $AB$. Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$.

$$\cos\angle AOB = \frac{\overrightarrow{OA}\cdot\overrightarrow{OB}}{\|\overrightarrow{OA}\|\|\overrightarrow{OB}\|} = \frac{\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}\cdot\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}}{\left\|\begin{pmatrix} 2 \\ 9 \\ -6 \end{pmatrix}\right\|\left\|\begin{pmatrix} 6 \\ 3 \\ 6 \end{pmatrix}\right\|} = \frac{12+27-36}{\sqrt{121}\cdot\sqrt{81}} = \frac{3}{99} = \frac{1}{33}$$

$$\implies \angle AOB = \cos^{-1}\left(\frac{1}{33}\right) \approx 1.5403 \text{ radians } \approx 88.3^\circ.$$